

GAME TEST REPORT

SlotForge One

Pre-Certification Self-Assessment

ISSUES FOUND

Report Number	BS-SLOTFORGE-20260731-144CD595
Date of Report	2026-07-31
Issued By	BlueStuff Game Studio · Automated Test Engine
Report Recipient	Smokestack
Engine Under Test	slotforge
Standard Tested Against	GLI-19 v3.0 — Interactive Gaming Systems
Classification	Pre-certification self-assessment (Non-Jurisdictional)
Software Fingerprint	sha256:b7861063fdb9bc4fb98a0a18c7253564f1fd967b... · 34 files
Result Summary	2 passed · 1 failed · 1 warnings · 0 manual · 0 source findings

Disclaimer. This is an automated pre-certification self-assessment produced by BlueStuff Game Studio's own test engine, published unmodified — including any failures. It is tested *against* the GLI-19 v3.0 technical standard but is **not** an accredited-laboratory certification and does not imply affiliation with, or endorsement by, Gaming Laboratories International. Final certification requires an independent accredited testing laboratory.

1. Executive Summary

This build did not pass every check. Details and reproduction steps are below.

2. Scope & Evaluation Elements

The following elements were evaluated to the extent applicable to automated testing of this software:

Element	Status
Software & System Version Control	Performed
Source Code Review	Performed
Game Accounting (RTP / payout)	Performed
RNG & Statistical Analysis (GLI-19 §3.2)	Performed
Functionality Testing	Partial

3. Requirement Results

Requirement	Result	Sample	Statistic
3.2.2	FAIL	240,000	p = 2.79e-5
3.2.3	PASS	240,000	p = 9.28e-1
3.2.4	PASS	180,000	p = 7.09e-1
4.7.1	WARN	120,000	—

4. Requirement Detail

3.2.2 FAIL

Shared RNG source (crypto-backed, used by every engine on this template). float() scaled to 32 bins, 2 independent draws combined by Fisher's method. 7 of the 7 tests named in §3.2.2 applied. 1 rejected at the 99% family-wise confidence level: duplicates (p = 2.788e-5, threshold 1.429e-3).

3.2.3 PASS

RNG output is uniformly distributed (float 32-bin: Fisher chi-square = 0.88, p = 0.9275; int 64-range: p = 0.5768).

3.2.4 PASS

Independence of the shared RNG (serial, overlaps, interplay) and of each engine's per-round outcomes (slotforge-mvp). 4 of the 7 tests named in §3.2.2 applied. None rejected at the 99% family-wise confidence level (Holm–Bonferroni; Fisher combined p = 9.989e-1).

4.7.1 WARN

Return-to-player measured per registered engine by driving play() with the template RNG: • example: no payouts observed over 60,000 spins — stub or non-payout engine; nothing to certify. • slotforge-mvp: RTP 94.099% ±8.915pp over 60,000 spins (hit 34.89%, max 385.0x) — provisional (widen sample). GLI-19 §4.7.1 requires a theoretical payout of at least 75%. (1 non-payout/stub engine(s) noted above.)

5. Source Code Review

No adversarial findings. The probes that were run did not surface an exploitable issue.

6. Software & System Version Control

A SHA-256 checksum was generated for each of the 34 source files in the engine under test. The aggregate fingerprint binds this report to exactly the code tested.

Aggregate fingerprint: sha256:b7861063fdb9bc4fb98a0a18c7253564f1fd967bcd25db790cdfd7d17235caf6

File	Bytes	SHA-256
mock-rgs/server.js	6,941	efd3b571471ff8715098...
package-lock.json	39,470	80396a8b9dc627a8b7bb...
package.json	650	dbe06c33c23778839840...
src/adapters/InlineWallet.js	3,591	0f3d448ea065d1965b13...
src/adapters/RgsAdapter.js	11,741	bf249f586850cb72fe3d...
src/adapters/WalletAdapter.js	1,827	96abb0fa90468bbcb94a...
src/app.js	2,960	4853c5de79c0d57c1eb9...
src/config/index.js	2,048	70ca00b00159577b15f3...
src/engines/GameEngine.js	3,261	83d6fff9314cb1845d45...
src/engines/example/ExampleEngine.js	2,827	b146f8322209c3764e3c...
src/engines/index.js	1,548	0b61a5e98c81add4284c...
src/engines/slotforge-mvp/SlotForgeEngine.js	19,265	d1e0ebee4f7d106838d1...
src/engines/slotforge-mvp/analytics.js	7,427	b709c62a10f07eee7b5b...
src/engines/slotforge-mvp/defaultMath.js	17,834	6873b7255c0b10641b26...
src/middleware/errorHandler.js	936	e3c1912b7fe62a86a9cd...
src/models/index.js	4,175	763a60ca94e6cec820e5...
src/routes/config.js	957	a4c5adc3cd5cceaabe52...
src/routes/game.js	1,742	07682a3da73b014775f1...
src/routes/session.js	1,309	f3d8ea81cc657ab65445...
src/server.js	1,010	cda1ea92956411d4dd94...
src/services/AuditService.js	1,205	2887a68af4ba2500ecb7...
src/services/GameService.js	13,281	2111fc9a04afb7c07507...
src/services/SessionManager.js	3,501	9a133289e6702199ceb7...
src/store/index.js	554	90e0279b64368867bc05...
src/store/memoryStore.js	1,524	33fcc249eed7130402f9...
src/store/mongoStore.js	1,177	89d67140ee3156976ce5...
src/utils/errors.js	1,710	0d4ee388580860da1aba...

src/utils/ids.js	1,036	19816c83e246299f080e...
src/utils/logger.js	1,938	6ab4fc4d1af74e62a7a4...
src/utils/money.js	996	6b633695f0ace232e6cd...
src/utils/rng.js	1,106	6a75832219a1d9540812...
tests/lifecycle.test.js	11,972	e4c304aebdb95a95b6f1...
tests/rgs-adapter.test.js	8,944	d75b50b824c88f8ab9e7...
tests/slotforge-mvp.test.js	44,552	e352794f8295f43f63be...

7. Verification & Reproduction

The test engine has no dependencies and no LLM in its numeric path — every statistic is computed by reviewable in-tree code. This run reproduces with:

```
node src/cli.ts --engine=slotforge --repo=D:\Projects\smokestack-original-games\BackendEngines\slotrocket-5c5a --samples=40000 --spins=60000 --replicates=2 --seed=pdf
```

The seven statistical tests are the ones GLI-19 §3.2.2 names explicitly (chi-square, overlaps, coupon collector, runs, interplay correlation, serial correlation, duplicates), evaluated collectively at 99% family-wise confidence and combined across independent replicate seeds.

8. Test Environment

Test engine	SportsITDev/gli-audit-engine
Runtime	Node v24.14.0
Seed	pdf
Started	2026-07-31T09:44:10.918Z
Finished	2026-07-31T09:44:12.780Z
Generated by	gli-audit-engine game-tester agent